

Chapter 2: Review of Probability and Statistics

By-Hand Exercises

1. Random variables and their realizations

An online shop records the following variables for each of its customers:

- (i) time spent on the website (in minutes),
 - (ii) number of items in the shopping basket,
 - (iii) payment method $\in \{\text{credit card}, \text{PayPal}, \text{invoice}\}$,
 - (iv) customer's body height (in cm),
 - (v) whether the customer returns at least one item $\in \{\text{yes}, \text{no}\}$.
- (a) Decide for each variable whether it is a discrete or a continuous random variable.
 - (b) For which of the variables does it make sense to compute a mean?
 - (c) Which tool describes the distribution of variable (i): a density function (pdf) or a probability mass function (pmf)? Which tool describes the distribution of variable (ii)? Which tool can be used for *both* (i) and (ii)?
 - (d) Let T_i denote the time that customer i spends on the website. Explain the difference between T_i and $T_{i,\text{obs}}$. Which of the two is a fixed number?

2. Probability mass function, mean and variance

Let W denote the number of customer complaints that a small company receives on a randomly chosen day. W has the following probability mass function:

w_k	0	1	2	3	4
$p_W(w_k)$	0.10	0.25	0.30	c	0.15

- (a) Determine c such that p_W is a valid probability mass function.
- (b) Compute $P(W \geq 2)$ and $P(W \leq 1)$.
- (c) Compute the expected number of complaints per day, $E(W)$.
- (d) Compute the variance $\text{Var}(W)$ and the standard deviation $\text{SD}(W)$.
- (e) Each complaint costs the company 20 EUR. In addition, the complaint hotline has fixed costs of 150 EUR per day. Thus the daily costs are $K = 150 + 20W$. Compute $E(K)$ and $\text{SD}(K)$.

3. Density function

The delay X (in hours) of a parcel delivery is a continuous random variable with density function

$$f_X(x) = \begin{cases} c \cdot x & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise,} \end{cases}$$

where $c > 0$ is a constant.

- (a) Determine c such that f_X is a valid density function.
- (b) Compute the probability that the delay is between 30 minutes and 90 minutes, i.e., $P(0.5 \leq X \leq 1.5)$.
- (c) Derive the cumulative distribution function $F_X(x) = P(X \leq x)$ for all $x \in \mathbb{R}$. Use it to compute $P(X > 1)$.
- (d) Compute the expected delay $E(X)$.

- (e) Compute $\text{Var}(X)$ and $\text{SD}(X)$.

Hint: It is often easier to use $\text{Var}(X) = E(X^2) - (E(X))^2$, where $E(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x) dx$.

- (f) Another delay V (in hours) has the density $f_V(v) = 4$ for $0 \leq v \leq 0.25$ and $f_V(v) = 0$ otherwise. Is this a valid density, even though $f_V(v) > 1$? Explain.

4. Empirical cumulative distribution function

A mail-order company records the delivery times (in days) of $n = 8$ randomly selected orders:

$$X_{1,\text{obs}}, \dots, X_{8,\text{obs}} = 7, 1, 5, 5, 2, 2, 2, 5.$$

- Compute the empirical cdf $\hat{F}_{X,\text{obs}}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_{i,\text{obs}} \leq x)$ at the points $x = 0, 2, 3.5, 6, 10$.
- Sketch the empirical cdf $\hat{F}_{X,\text{obs}}(x)$ for $x \in [0, 8]$. Where does it jump, and how large are the jumps?
- Use the empirical cdf to estimate the probability that a delivery takes more than 3 days.
- Estimate the probability mass function $p_X(k)$, $k = 1, \dots, 7$, by relative frequencies.

5. Calculation rules for mean and variance

The monthly sales X (in units) of a product are random with $E(X) = 600$ and $\text{SD}(X) = 40$. Each unit is sold for 15 EUR, and the fixed costs are 3000 EUR per month. The monthly profit is thus

$$\Pi = 15X - 3000.$$

- Compute the expected monthly profit $E(\Pi)$.
- Compute $\text{Var}(\Pi)$ and $\text{SD}(\Pi)$. Why do the fixed costs not matter here?
- The company sells a second product with monthly sales Y and $E(Y) = 250$. What is the expected total number of units sold, $E(X + Y)$? Do you need to assume that X and Y are independent?
- The daily maximum temperature C (in degrees Celsius) in July has mean $E(C) = 20$ and standard deviation $\text{SD}(C) = 3$. In degrees Fahrenheit, the temperature is $F = 1.8C + 32$. Compute $E(F)$ and $\text{SD}(F)$.
- Let X be any random variable with mean μ_X and standard deviation $\sigma_X > 0$. Show that the standardized random variable $Z = \frac{X - \mu_X}{\sigma_X}$ has $E(Z) = 0$ and $\text{Var}(Z) = 1$.

6. Sample mean and sample variance

A bakery records its daily revenue (in 1000 EUR) on $n = 6$ randomly selected days:

$$X_{1,\text{obs}}, \dots, X_{6,\text{obs}} = 11, 9, 11, 6, 8, 9.$$

- Compute the sample mean $\bar{X}_{\text{obs}}$ by hand.
- Compute the sample variance $\hat{\sigma}_{X,\text{obs}}^2$ and the sample standard deviation $\hat{\sigma}_{X,\text{obs}}$ by hand.
- The bakery wants to report the revenue in EUR instead of 1000 EUR. How do the sample mean, the sample variance and the sample standard deviation change?
- Is $\bar{X}_{\text{obs}}$ equal to the true mean $E(X)$ of the daily revenue? Explain.

7. Covariance and correlation

A retail chain records the advertising expenditures X (in 1000 EUR) and the sales Y (in 1000 units) of $n = 5$ of its stores:

Store i	1	2	3	4	5
$X_{i,\text{obs}}$	5	6	8	7	4
$Y_{i,\text{obs}}$	17	18	24	20	16

- Compute the sample covariance $\widehat{\text{Cov}}_{\text{obs}}(X, Y)$ by hand.
- Compute the sample correlation $\widehat{\text{Cor}}_{\text{obs}}(X, Y)$ by hand and interpret it.
- Suppose the advertising expenditures are recorded in EUR (instead of 1000 EUR), and the sales in units (instead of 1000 units). How do the covariance and the correlation change?
Hint: $\text{Cov}(a_X X + b_X, a_Y Y + b_Y) = a_X a_Y \text{Cov}(X, Y)$.
- Does the result in (b) show that advertising *causes* higher sales? Explain.

8. Joint, marginal and conditional distributions

An online shop classifies its customers by $X \in \{0, 1\}$, where $X = 0$ denotes a new customer and $X = 1$ a returning customer. Let $Y \in \{1, 2, 3\}$ denote the number of items a customer buys. The joint probability mass function $p_{Y,X}(y, x) = P(Y = y, X = x)$ is given by

	$Y = 1$	$Y = 2$	$Y = 3$
$X = 0$	0.20	0.15	0.05
$X = 1$	0.10	0.25	0.25

For discrete random variables, all formulas for densities carry over with sums instead of integrals. For instance, the conditional pmf is $p_{Y|X}(y | x) = p_{Y,X}(y, x) / p_X(x)$.

- Check that $p_{Y,X}$ is a valid joint probability mass function.
- Compute the marginal pmfs $p_X(x)$ and $p_Y(y)$.
- Compute the conditional pmf of Y given $X = 0$, and the conditional pmf of Y given $X = 1$.
- Compute the conditional means $E(Y | X = 0)$ and $E(Y | X = 1)$, as well as the unconditional mean $E(Y)$. Interpret the results.
- Are Y and X independent? Justify your answer.

9. The normal distribution

A machine fills coffee into cups. The filled amount X (in ml) is normally distributed, $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = 200$ and $\sigma = 5$. Let $\Phi(z) = P(Z \leq z)$ denote the cdf of the standard normal distribution $Z \sim \mathcal{N}(0, 1)$. Use a table of Φ and the following quantiles: $\Phi(-1.645) = 0.05$ and $\Phi(-2.326) = 0.01$.

- Compute the probability that a cup contains less than 188 ml.
- Compute the probability that a cup contains between 195 ml and 205 ml.
Hint: By symmetry, $\Phi(-z) = 1 - \Phi(z)$.
- Which amount is undercut by only 5% of all cups? That is, find x such that $P(X \leq x) = 0.05$.
- The company wants that at most 1% of all cups contain less than 188 ml. The standard deviation $\sigma = 5$ cannot be changed, but the machine's mean μ can be adjusted. How large must μ be at least?

10. The data-generating process

The monthly electricity costs Y_i (in EUR) and the living space X_i (in m^2) of households $i = 1, \dots, n$ are generated by the DGP

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

with $\beta_1 = 40$, $\beta_2 = 0.8$, $\sigma = 10$, and $X_i \sim \mathcal{N}(80, 20^2)$ independent of ε_i .

- (a) Which variable is the outcome and which is the predictor? What does the error term ε_i capture? Interpret β_2 .
- (b) Compute the values of the regression function $m(x) = E(Y_i | X_i = x)$ at $x = 50$ and $x = 100$.
- (c) Which distribution does Y_i have, given that $X_i = 100$? Compute the probability that a household with 100 m² has monthly electricity costs above 140 EUR. (Use a table of the standard normal cdf Φ .)
- (d) In practice, which parts of this DGP are usually unknown?
- (e) The DGP assumes i.i.d. sampling. What does this mean here? Give an example of a data collection where the i.i.d. assumption would be violated.

11. Association is not causation

A kiosk next to an outdoor swimming pool records, for each day i , its ice cream sales X_i (in 100 portions) and the number of visitors Y_i of the swimming pool (in 100 persons). Consider the following DGP, where Z_i is the daily maximum temperature (in degrees Celsius):

$$\begin{aligned} Z_i &\sim \mathcal{N}(22, 5^2), \\ X_i &= 2 + 0.5 Z_i + u_i, & u_i &\sim \mathcal{N}(0, 1), \\ Y_i &= 1 + 0.3 Z_i + v_i, & v_i &\sim \mathcal{N}(0, 1), \end{aligned}$$

with Z_i , u_i and v_i independent of each other.

- (a) Does the number of pool visitors Y_i depend on the ice cream sales X_i in this DGP?
- (b) Nevertheless, data from this DGP show a strong positive correlation between X_i and Y_i . Explain why. What is the role of Z_i ?
- (c) Suppose we only use days with (almost) the same maximum temperature, e.g., between 24 and 26 degrees Celsius. What correlation between X_i and Y_i do you expect for these days? Explain.
- (d) Compute $E(X_i)$ and $E(Y_i)$.
- (e) Companies that spend a lot of money on management consultants have, on average, lower profits than other companies. Does this show that consultants are harmful? Give an alternative explanation.